

SIMATS ENGINEERING

ASSESSMENT TOOL 2

COURSE NAME: INTERNET PROGRAMMING

COURSE CODE: CSA43

COURSE OUTCOME COVERED

CO-1: Develop well-structured, standards-compliant web pages using HTML/XHTML and XML, and apply Internet protocols and HTTP communication mechanisms for web content delivery. (L2)

Assessment Tool : Data Interpretation

Weightage: 20%

QUESTIONS:

Total Marks: 20

1: HTTP Response Analysis

A web server returns the following HTTP response header:

HTTP/1.1 200 OK

Date: Mon, 01 Apr 2026 10:00:00 GMT

Content-Type: text/html

Content-Length: -150

Connection: keep-alive

Questions:

1. Identify any incorrect or invalid fields in the header.
2. Analyze the issue with `Content-Length` and explain its impact.
3. Determine whether the status code matches the response correctness.
4. Suggest corrections for the identified errors.

2: DOM Structure Analysis

Consider the following DOM tree representation:

```

<html>
  <head>
    <title>Test Page</title>
  </head>
<body>
  <div>
    <p>Hello World
  </div>
</body>
</html>

```

Questions:

1. Identify missing or improperly closed elements.
2. Analyze how the browser will interpret this structure.
3. Determine the impact on rendering and layout.
4. Suggest corrections to make the DOM valid.

3: GET vs POST in Login System

A login system uses two different methods:

Method	URL Example	Data Visibility	Security Level
GET	/login?user=admin&pass=1234	Visible in URL	Low
POST	/login	Hidden in body	Moderate

Questions:

1. Interpret the differences in data transmission between GET and POST.
2. Analyze which method is more suitable for login and why.
3. Identify security risks in using GET for authentication.
4. Suggest best practices for secure login implementation.

4: HTML Code Inspection

Analyze the following HTML snippet:

```

<!DOCTYPE html>
<html>
<head>
<title>Sample</title>
</head>
<body>
<h1>Welcome
<p>This is a paragraph

```

```

</body>
</html>
```

Questions:

1. Identify missing closing tags.
2. Analyze issues with the `<img>` tag.
3. Determine how browsers will handle these errors.
4. Suggest a corrected version of the code.

5: Browser-Server Communication Flow

The following sequence describes a web request:

1. User enters a URL in the browser
2. DNS lookup occurs
3. HTTP request is sent
4. Server processes request
5. Response is returned
6. Browser renders page

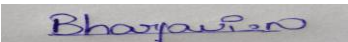
Questions:

1. Interpret each step in the communication flow.
2. Identify where delays can occur.
3. Analyze the role of DNS in the process.
4. Suggest optimizations to improve performance.

Code of Conduct:

*I **Bhargavi. N (192411144)** certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.
I understand that any violation of academic integrity rules will result in disciplinary action.*

Signature of the student:



Web Technologies Analysis Assignment

1. HTTP Response Analysis

```
HTTP/1.1 200 OK
Date: Mon, 01 Apr 2026 10:00:00 GMT
Content-Type: text/html
Content-Length: -150
Connection: keep-alive
```

- Invalid Field: Content-Length = -150
- Impact: Browser cannot determine response size.
- Status Code: 200 OK is valid.
- Correction: Content-Length: 150

2. DOM Structure Analysis

```
<html>
  <head>
    <title>Test Page</title>
  </head>
  <body>
    <div>
      <p>Hello World
    </div>
  </body>
</html>
```

Error: Missing </p> tag.

Browser Behavior: Browser automatically closes the paragraph.

Correct Version

```
<div>
  <p>Hello World</p>
</div>
```

3. GET vs POST Analysis

Feature	GET	POST
Data Location	URL	Request Body
Visibility	Visible	Hidden
Security	Low	Moderate

Best Method: POST

- Use HTTPS.
- Hash passwords.
- Use sessions and tokens.

4. HTML Code Inspection

```
<h1>Welcome
<p>This is a paragraph

```

- Missing </h1>
- Missing </p>
- Missing alt attribute in image tag

Correct Version

```
<h1>Welcome</h1>
<p>This is a paragraph</p>

```

5. Browser-Server Communication Flow

```
User enters URL
  ↓
DNS Lookup
  ↓
HTTP Request
  ↓
Server Processing
  ↓
HTTP Response
  ↓
Browser Renders Page
```

Possible Delays

- DNS lookup time
- Slow internet connection
- Server overload
- Large images and scripts

Optimizations

- Use CDN
- Enable caching
- Compress images
- Minify CSS and JavaScript
- Use HTTP/2 and HTTPS

RUBRICS FOR CALCULATION:

Criteria	Weightage	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)
Understanding of Data	25%	Accurately interprets all data patterns, trends, and relationships	Interprets data correctly with minor gaps	Partial understanding; misses key trends	Misinterprets or fails to understand data
Analysis	25%	Provides deep analysis with strong logical reasoning and justification	Provides reasonable analysis with some justification	Basic analysis with weak reasoning	No meaningful analysis provided
Application of Concepts	20%	Effectively applies relevant concepts/tools to interpret data accurately	Applies concepts with minor errors	Limited application; requires guidance	Unable to apply concepts to data
Conclusion	20%	Draws clear, meaningful conclusions with strong insights and implications	Provides valid conclusions with moderate insights	Conclusions are vague or weak	No clear or valid conclusions
Clarity	10%	Presents interpretation clearly using proper structure, visuals, and terminology	Generally clear with minor issues	Lacks clarity or proper structure	Poor presentation and difficult to understand